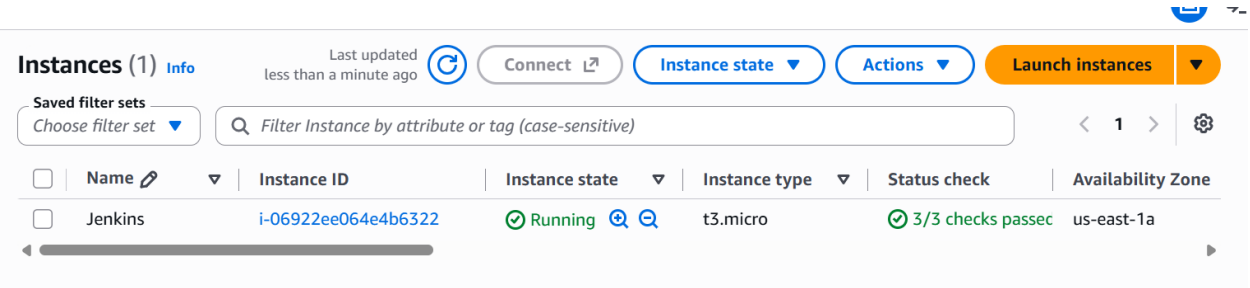
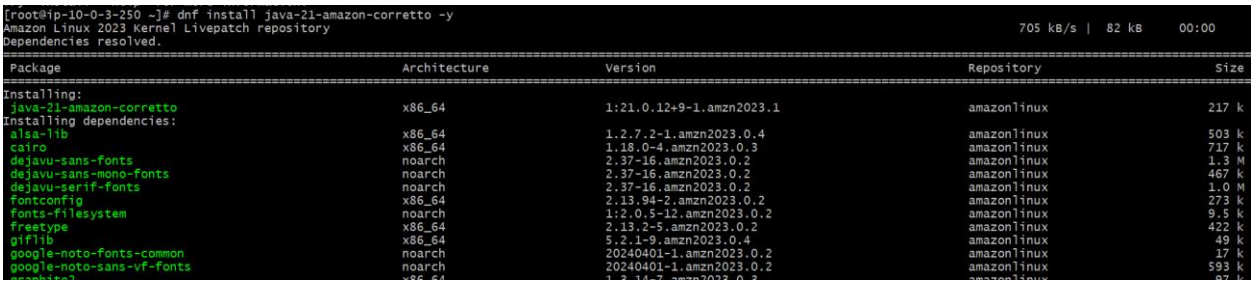
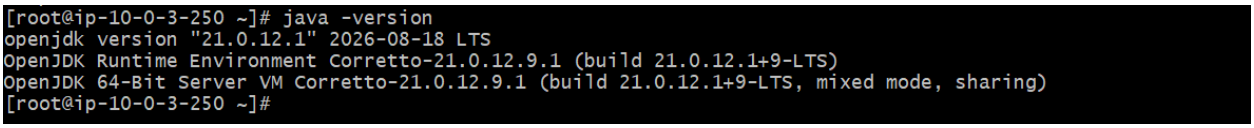
Install Jenkins and run Jenkins on port number 8081.



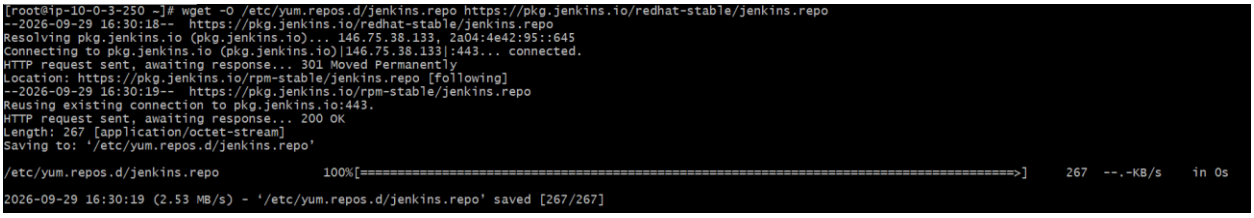
Create an instance, by allowing the port 8081 and enable the public ip and using ssh conect to terminal



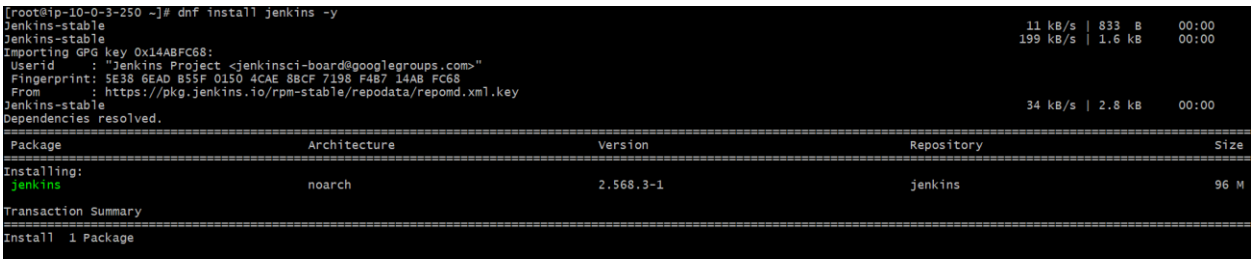
Install java



Check the version of java



Download the jenkins repository



Install jenkins

```
[root@ip-10-0-3-250 ~]# cat /usr/lib/systemd/system/jenkins.service
#
# This file is managed by systemd(1). Do NOT edit this file manually!
# To override these settings, run:
#
#     systemctl edit jenkins
#
# For more information about drop-in files, see:
#
#     https://www.freedesktop.org/software/systemd/man/systemd.unit.html
#
```

Cat the service file and change the port from 8080 to 8081

```
# Port to listen on for HTTP requests. Set to -1 to disable.
# To be able to listen on privileged ports (port numbers less than 1024),
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
Environment="JENKINS_PORT=8081"
```





This site can't be reached

Port 8081 is not responding

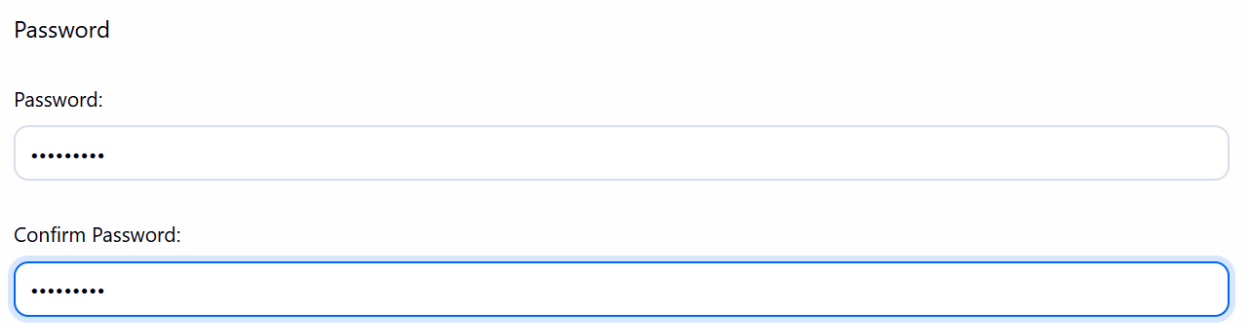
```
[root@ip-10-0-3-250 ~]# ss -lntp | grep java
LISTEN 0      50                 *:8080            *:*    users:(("java",pid=70767,fd=9))
[root@ip-10-0-3-250 ~]# systemctl daemon-reload
[root@ip-10-0-3-250 ~]# systemctl restart jenkins
[root@ip-10-0-3-250 ~]# ss -lntp | grep java
LISTEN 0      50                 *:8081            *:*    users:(("java",pid=71100,fd=9))
[root@ip-10-0-3-250 ~]#
```



Secure Jenkins server

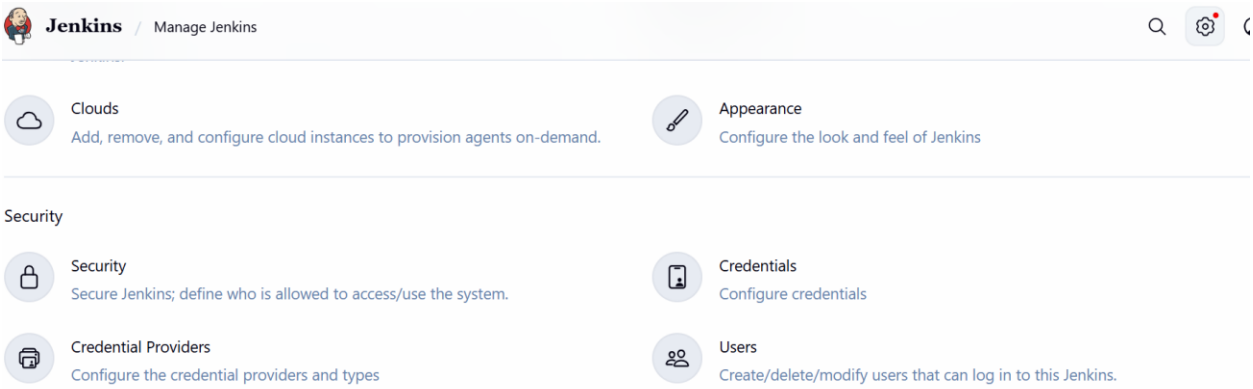
```
[root@ip-10-0-3-250 ~]# /var/lib/jenkins/secrets/initialAdminPassword
-bash: /var/lib/jenkins/secrets/initialAdminPassword: Permission denied
[root@ip-10-0-3-250 ~]# cat /var/lib/jenkins/secrets/initialAdminPassword
cf6ec2cbf36745678be1e95e91d0fcf4
```

Cat the location of secrets for the password

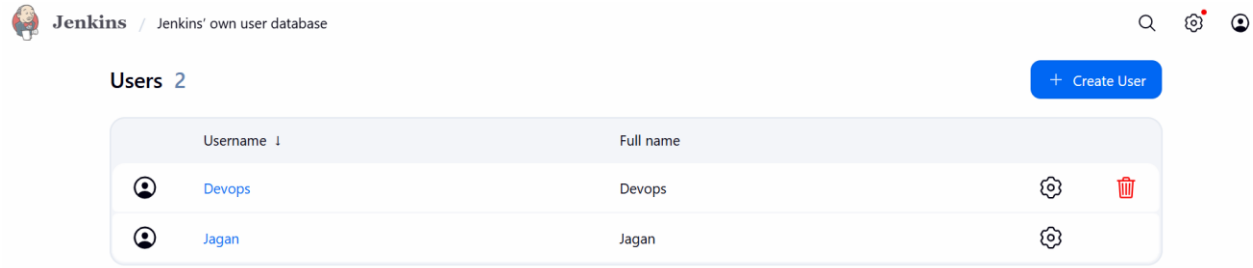


To secure jenkins change the password and delete the generated or actual password

Create users called DevOps, Testing in Jenkins with Limited access.



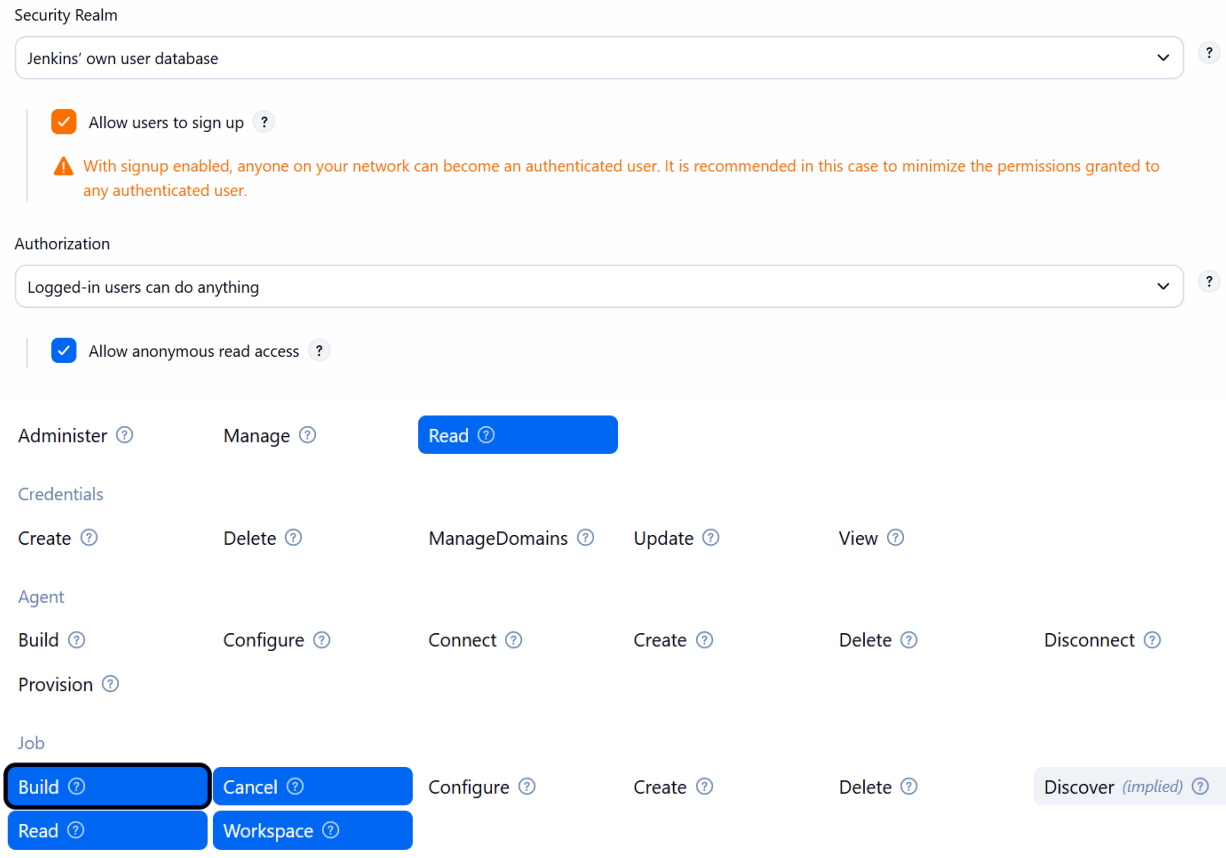
Click on the users



Created a user Devops



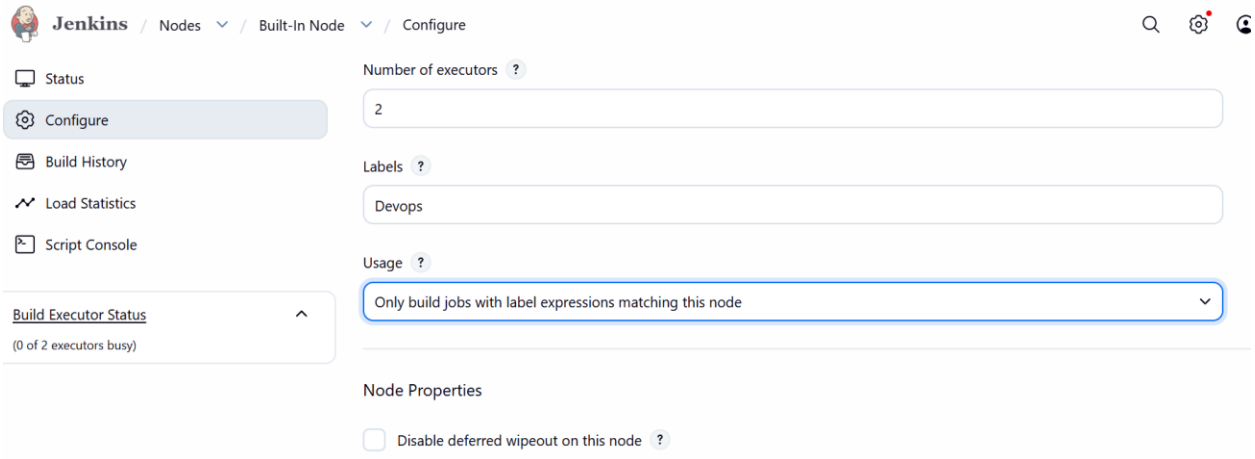
Created another user



for Devops we'll give few more access and for Testing we give only read access



Configure labels and restrict the jobs to execute based on label only.



Create Three sample jobs using the below URL

New Item

Enter an item name

Job1

Select an item type

Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell

Command

See the list of available environment variables

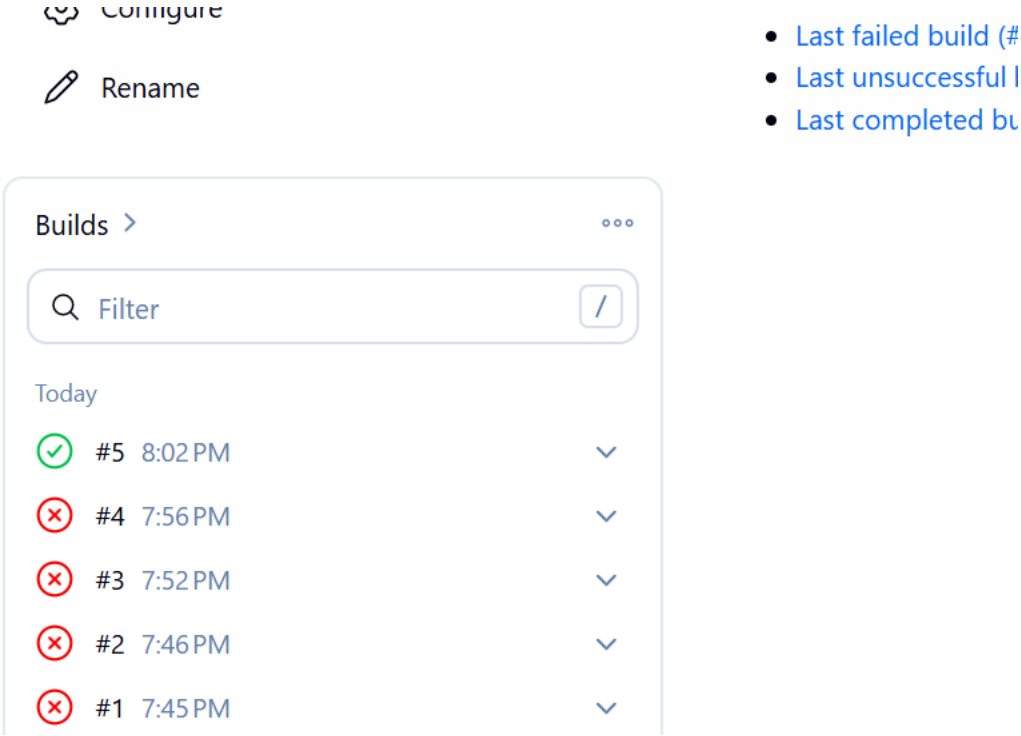
```
echo "Hello from Jenkins Job 1"
```

Builds

Pending

#1

'Jenkins' is reserved for jobs with matching label expression



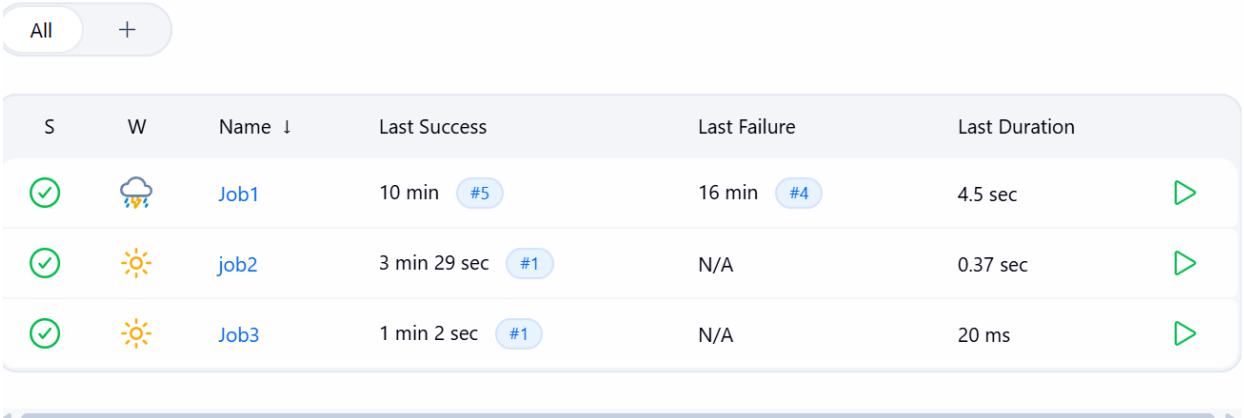
Job 1 is successfull

```
-rw-r--r--. 1 jenkins jenkins 5830 Sep 29 20:09 pom.xml
drwxr-xr-x. 4 jenkins jenkins   30 Sep 29 20:09 src
Finished: SUCCESS
```

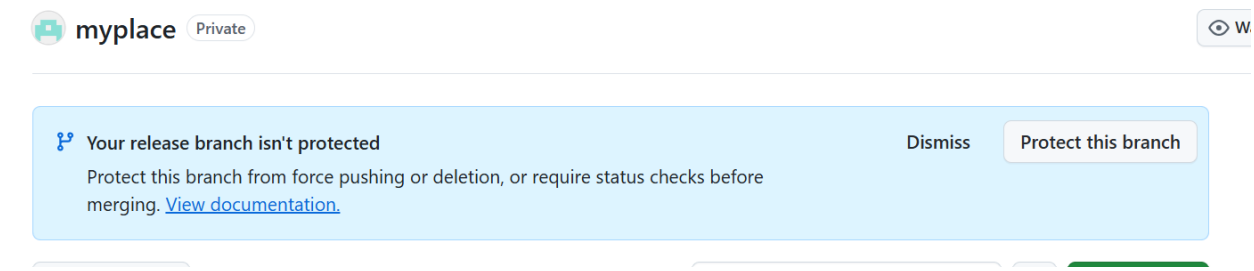
Job 2 is also success

```
Started by user Jagan
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Job3
[Job3] $ /bin/sh -xe /tmp/jenkins17906888299550576842.sh
+ echo 'Job3 is running'
Job3 is running
++ hostname
+ echo 'Hostname: ip-10-0-3-250.ec2.internal'
Hostname: ip-10-0-3-250.ec2.internal
+ echo 'Workspace: /var/lib/jenkins/workspace/Job3'
Workspace: /var/lib/jenkins/workspace/Job3
+ ls -la
total 0
drwxr-xr-x. 2 jenkins jenkins  6 Sep 29 20:12 .
drwxr-xr-x. 5 jenkins jenkins 42 Sep 29 20:12 ..
Finished: SUCCESS
```

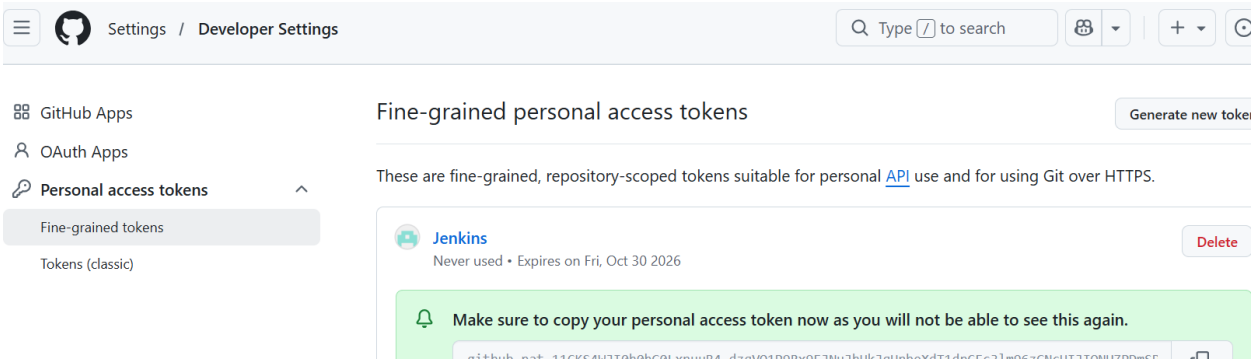
Job 3 also success

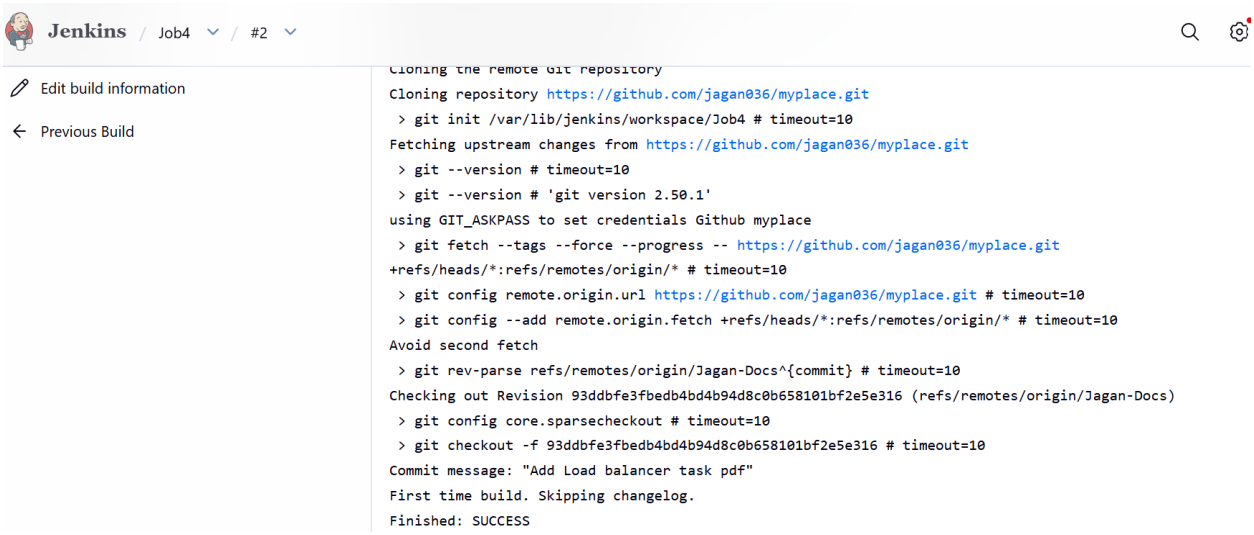


Create one Jenkins job using GitHub Private repository.



Make it as private





Jenkins Job4

↓

GitHub credentials

(jagan036 + PAT)

↓

Private repository

jagan036/myplace 🔒

↓

Jenkins clones the code

↓

Build